🗄️ INSTRUCTOR-LED COURSE SYLLABUS · MYSQL

MySQL Foundations

An instructor-led course in 14 three-hour sessions, in two parts. Install MySQL, build and query a real database, join and summarize its tables, then go beyond the basics (window functions, indexes, design, transactions, security and backups) and finish with a database you design yourself. Part 1 (7 sessions) also runs on its own.

Course site: rays-mysql.netlify.app

`14 sessions × 3 h` `42 class hours` `25 lessons` `2–3 h homework per session`
`Part 1 alone: 7 sessions` `MySQL 9.7 LTS`

📚 COURSE DESCRIPTION

Every website that remembers you keeps its data in a database, and most of them speak SQL. This course teaches MySQL 9.7 LTS from the first install to a database you design, build and defend yourself: creating tables with the right types and rules, asking precise questions with SELECT, connecting tables with keys and joins, and then the tools working developers and analysts use every week.

Every session uses the same database: an online **bookstore** with real classic books and fictional customers, orders and staff. It grows with you, from three empty tables in Session 2 to a relational database with orders, order items and an employee hierarchy, so every new idea is practiced on data you already know.

🎯 By the end you will be able to

- Install MySQL 9.7, secure it, and diagnose connection errors by their numbers
- Create databases and tables with fitting data types and constraints, and insert, update and delete rows safely
- Filter, sort, page, summarize and nest queries, including GROUP BY and subqueries
- Design related tables with primary and foreign keys, and combine them with joins
- Write CTEs, recursive queries and window functions for rankings and running totals
- Speed up queries with indexes, simplify them with views, and normalize a design
- Use stored procedures, transactions, JSON columns, accounts and permissions, imports, exports and backups
- Design, build and present a complete database of your own

Part 1 only: Sessions 1 to 7 (21 class hours, 13 lessons) run as a complete short course: from installing MySQL to joining tables, ending with the Part 1 assessment. Part 2 (Sessions 8 to 14) can follow later, or run on its own for learners who already have the Part 1 skills.

✅ PREREQUISITES

- No database or programming experience. Comfortable with files and folders, and with a spreadsheet (filtering, sorting, a simple formula).
- A Windows, macOS or Linux laptop on which you can install software (or run Docker), with about 1.5 GB free. MySQL 9.7 is installed at home before Session 1, following Lesson 1.2.

- Starting with Part 2 only: Part 1 or equivalent experience (SELECT with WHERE, ORDER BY and GROUP BY, primary and foreign keys, joins, and the course's bookstore database).

💡 FORMAT

14 sessions × 3 hours = 42 class hours, two lessons per session (Sessions 7 and 13 teach one lesson and give the rest of the time to an assessment or the final-project studio), in two parts: **Part 1** (Sessions 1 to 7) and **Part 2** (Sessions 8 to 14). Every lesson on the course site is written for a solo learner and takes about 75 to 110 minutes, so two lessons do not fit whole in one room session. The room protects what is hardest alone: the **mental model**, a **live demo** you type along with, a **guided lab** with help on hand, and **starting** each lesson project. At home you finish the projects and do each lesson's exercises, quiz and journal entry on the free course site: about **two to three hours** between sessions.

Each session runs 0:00 to 3:00: a warm-up on last session, the first lesson, a 10-minute break, the second lesson, and a wrap-up with an exit ticket and the next bring list. Single-lesson sessions (7, 13, 14) give the second half to the Part 1 assessment or the final-project studio.

> **Part 1 only:** Sessions 1 to 7 (21 class hours, 13 lessons) run as a complete short course: from installing MySQL to joining tables, ending with the Part 1 assessment. Part 2 (Sessions 8 to 14) can follow later, or run on its own for learners who already have the Part 1 skills.

Twice weekly: about 7 weeks for the full course (Part 1 alone: 4 weeks). Two homework sets a week, so plan on four to six hours outside class.

Weekly: 14 weeks (Part 1 alone: 7), with a full week for each homework set.

Intensive days: two sessions a day; homework becomes evening practice and the final project is finished afterward.

📖 SESSION BY SESSION

Lesson titles link to the lesson pages on the course site. Homework is the time to finish that session's lesson projects, exercises, quizzes and journal entries.

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
PART 1: SQL FOUNDATIONS · SESSIONS 1 TO 7			
1 **What Databases Are and Installing MySQL** SETUP	1.1 What Is a Database? 1.2 Installing MySQL	Explain what a database and a DBMS are and why tables and keys beat a spreadsheet, run first SQL statements, and have MySQL 9.7 running, secured and verified on your own laptop.	2–3 h
2 **The MySQL Client and Your First Tables** SETUP TABLES & DATA	1.3 MySQL CLI & phpMyAdmin 2.1 Creating Databases & Tables	Move around confidently in the mysql client (and phpMyAdmin), run .sql files, and create the bookstore database and its first tables.	2–3 h

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
3 **Data Types, Constraints and CRUD** TABLES & DATA	2.2 Data Types & Constraints 2.3 CRUD Operations	Choose data types and constraints that refuse bad data, and insert, read, update and delete rows safely.	2–3 h
4 **Filtering, Sorting and Limiting** QUERYING	3.1 WHERE Clauses, Operators & Filtering 3.2 Sorting & Limiting Results	Filter rows with WHERE, operators, LIKE, IN and NULL tests, and sort and page results. Includes Checkpoint 1.	2–3 h
5 **Aggregates and Subqueries** QUERYING	3.3 Aggregate Functions & GROUP BY 3.4 Subqueries & Nested Queries	Summarize data with COUNT, SUM, AVG, GROUP BY and HAVING, and nest queries inside queries.	2–3 h
6 **Relationships and Foreign Keys** RELATIONSHIPS	4.1 Table Relationships 4.2 Foreign Keys & Referential Integrity	Turn a flat table into related tables, and protect relationships with foreign keys and their ON DELETE / ON UPDATE rules.	2–3 h
7 **Joins and the Part 1 Assessment** RELATIONSHIPS	4.3 Joins	Combine tables with inner, left, self and cross joins, then show what you learned in the Part 1 assessment.	2–3 h
PART 2: BEYOND THE BASICS · SESSIONS 8 TO 14			
8 **CTEs and Window Functions** MODERN SQL	5.1 CTEs & Recursive Queries 5.2 Window Functions	Structure queries with common table expressions, walk hierarchies with recursive CTEs, and rank and total rows with window functions.	2–3 h
9 **Indexes and Views** DESIGN & SPEED	6.1 Indexes & Query Optimization 6.2 Views	Read EXPLAIN, add the indexes a query needs, and package queries as views.	2–3 h
10 **Normalization and Stored Programs** DESIGN & SPEED ADVANCED FEATURES	6.3 Database Design & Normalization 7.1 Stored Procedures & Functions	Normalize a design to third normal form, and write stored procedures and functions.	2–3 h

SESSION	LESSONS	WHAT YOU WILL BE ABLE TO DO	HOMEWORK
11 **Transactions and JSON** ADVANCED FEATURES	7.2 Transactions 7.3 JSON Columns	Group changes into transactions that succeed or fail together, and store and query JSON columns. Includes Checkpoint 2.	2–3 h
12 **Users, Permissions and Backups** ADVANCED FEATURES FINAL PROJECT	7.4 User Management & Permissions 8.1 Importing & Exporting Data	Create accounts with least privilege and roles, import and export data, and back up and restore a database. The final-project brief is handed out.	2–3 h
13 **Final Project Kickoff and Studio I** FINAL PROJECT	8.2 Final Project: Design and Build a Database	Design the final project's database from a brief: entities, keys and rules, then build and load it, with milestone check-ins in the room.	3–5 h
14 **Final Project Studio II, Showcase and Next Steps** FINAL PROJECT	8.2 Final Project: Design and Build a Database 8.3 Next Steps & Best Practices	Finish the final project (queries, views, an account with the right permissions, a backup), present it, and plan your next steps with MySQL.	2–3 h

Total out-of-class time: about 29 hours or more across the course.

🕑 SESSION OUTLINES

Times run from the start of each session (0:00 to 3:00). Shift the block to any start time.

SESSION 1 What Databases Are and Installing MySQL

You build: 1.1 A paper design of the bookstore database this course builds; **1.2** A working MySQL 9.7 server and an installation health check you can rerun any time

0:00–0:30	Welcome, environment check and diagnostic
0:30–1:25	1.1 What Is a Database? frame and mental model, live demo, guided lab, start the lesson project
1:25–1:35	*Break*
1:35–2:50	1.2 Installing MySQL frame and mental model, install clinic, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 2 The MySQL Client and Your First Tables

You build: **1.3** explore.sql, a script that reports on your server, run three ways; **2.1** create_bookstore.sql, a script that builds the empty bookstore database from scratch

0:00–0:10	Warm-up
0:10–1:20	1.3 MySQL CLI & phpMyAdmin frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:50	2.1 Creating Databases & Tables frame and mental model, live demo, guided lab, build the lesson project
2:50–3:00	Wrap-up

SESSION 3 Data Types, Constraints and CRUD

You build: **2.2** employees.sql: the bookstore's staff table with five kinds of constraint, tested with good and bad rows; **2.3** load_bookstore_data.sql, which fills the bookstore with the 10 books, 8 customers and 5 categories that Module 3 queries

0:00–0:10	Warm-up
0:10–1:20	2.2 Data Types & Constraints frame and mental model, live demo, guided lab, start the lesson project
1:20–1:30	*Break*
1:30–2:50	2.3 CRUD Operations frame and mental model, live demo, guided lab, build the lesson project
2:50–3:00	Wrap-up

SESSION 4 Filtering, Sorting and Limiting

You build: **3.1** saved_searches.sql, six searches the bookstore staff can rerun any time; **3.2** storefront.sql, the queries behind four pages of the bookstore's website

> Checkpoint 1 (30 minutes): a short task on Modules 1 and 2.

0:00–0:10	Warm-up
0:10–1:15	3.1 WHERE Clauses, Operators & Filtering frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:20	3.2 Sorting & Limiting Results frame and mental model, live demo, guided lab, start the lesson project
2:20–2:50	**Checkpoint 1 (Modules 1 and 2)**
2:50–3:00	Wrap-up

SESSION 5 Aggregates and Subqueries

You build: 3.3 inventory_report.sql, a script that prints a five-part bookstore report; **3.4** price_review.sql, a five-part price review for the store owner

0:00–0:10	Warm-up
0:10–1:10	`3.3` Aggregate Functions & GROUP BY frame and mental model, live demo, guided lab, start the lesson project
1:10–1:20	*Break*
1:20–2:40	`3.4` Subqueries & Nested Queries frame and mental model, live demo, guided lab, start the lesson project
2:40–2:50	Error clinic
2:50–3:00	Wrap-up

SESSION 6 Relationships and Foreign Keys

You build: 4.1 the relational bookstore: authors, book_categories, orders, order_items and employees, the database every later lesson uses; **4.2** a book_reviews table with two different delete rules, and prove each rule works

0:00–0:10	Warm-up
0:10–1:15	`4.1` Table Relationships frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:40	`4.2` Foreign Keys & Referential Integrity frame and mental model, live demo, guided lab, start the lesson project
2:40–2:50	Relationship map
2:50–3:00	Wrap-up

SESSION 7 Joins and the Part 1 Assessment

You build: 4.3 sales_report.sql, a six-part sales and staff report built from joins

> Part 1 assessment and review. When Part 1 runs on its own, this session closes the course.

0:00–0:10	Warm-up
0:10–1:15	`4.3` Joins frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–1:35	Part 1 assessment: brief and setup
1:35–2:35	Part 1 assessment
2:35–2:50	Part 1 review
2:50–3:00	Wrap-up and close

SESSION 8 CTEs and Window Functions

You build: 5.1 org_report.sql, a staff and sales briefing: org chart, team sizes, a chain of command and a month-by-month sales calendar; **5.2** sales_dashboard.sql: running revenue, month-over-month change, a best-seller leaderboard and the top customer in every city

0:00–0:20	Welcome to Part 2: environment check and Part 1 refresher
0:20–1:15	`5.1` CTEs & Recursive Queries frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:35	`5.2` Window Functions frame and mental model, live demo, guided lab, start the lesson project
2:35–2:50	Debug clinic: both lessons
2:50–3:00	Wrap-up

SESSION 9 Indexes and Views

You build: **6.1** tune_sales.sql: three slow sales reports, measured, indexed and measured again; **6.2** reporting_views.sql: a rerunnable script that creates four reporting views for the store manager

0:00–0:10	Warm-up
0:10–1:10	`6.1` Indexes & Query Optimization frame and mental model, live demo, guided lab, start the lesson project
1:10–1:20	*Break*
1:20–2:25	`6.2` Views frame and mental model, live demo, guided lab, start the lesson project
2:25–2:50	Clinic: a view's plan, and the debug handouts
2:50–3:00	Wrap-up

SESSION 10 Normalization and Stored Programs

You build: **6.3** book_club.sql: a messy book-club sign-up sheet turned into four normalized tables, with a proof that nothing was lost; **7.1** bookstore_routines.sql: two functions and two procedures the bookstore's app can call

0:00–0:10	Warm-up
0:10–1:05	`6.3` Database Design & Normalization frame and mental model, live demo, guided lab, start the lesson project
1:05–1:15	*Break*
1:15–2:25	`7.1` Stored Procedures & Functions frame and mental model, live demo, guided lab, start the lesson project
2:25–2:50	Clinic: design review and the debug handouts
2:50–3:00	Wrap-up

SESSION 11 Transactions and JSON

You build: **7.2** checkout.sql: a checkout procedure that places an order and charges store credit, all or nothing; **7.3** catalog_feed.sql, a script that tags books, lists award winners and prints a JSON feed for the online store

> Checkpoint 2 (30 minutes): a short task on Modules 5 and 6.

0:00–0:10	Warm-up
0:10–0:40	**Checkpoint 2 (Modules 5–6)**
0:40–1:40	`7.2` Transactions frame and mental model, live demo, guided lab, start the lesson project
1:40–1:50	*Break*
1:50–2:50	`7.3` JSON Columns frame and mental model, live demo, guided lab, start the lesson project
2:50–3:00	Wrap-up

SESSION 12 Users, Permissions and Backups

You build: **7.4** team_access.sql, a rerunnable script that creates the bookstore team's roles and accounts; **8.1** a backup kit: a least-privilege backup account, a tested dump, a monthly sales CSV and a repeatable import script

> Final project brief: a database for the Maple Street Tool Library.

0:00–0:10	Warm-up
0:10–1:15	`7.4` User Management & Permissions frame and mental model, live demo, guided lab, start the lesson project
1:15–1:25	*Break*
1:25–2:30	`8.1` Importing & Exporting Data frame and mental model, live demo, guided lab, start the lesson project
2:30–2:50	**Final project brief**
2:50–3:00	Wrap-up

SESSION 13 Final Project Kickoff and Studio I

You build: **8.2** tool_library: six tables, a view, two procedures, two roles, a backup and a one-page write-up

> Studio: design and build, with milestone check-ins.

0:00–0:15	Warm-up
0:15–1:20	8.2 Final Project: Design and Build a Database kickoff: the brief, deliverables and rubric, frame and live demo, studio: Milestone 1, Design
1:20–1:30	*Break*
1:30–2:50	8.2 Final Project: Design and Build a Database guided lab, studio: Milestone 2, Build the schema, studio: Milestone 3, Test data
2:50–3:00	Wrap-up

SESSION 14 Final Project Studio II, Showcase and Next Steps

You build: **8.2** tool_library: six tables, a view, two procedures, two roles, a backup and a one-page write-up; **8.3** health_check.sql, a reusable audit for any database, and a written learning plan

> Studio, then a showcase of every learner's project.

0:00–0:10	Warm-up
0:10–1:30	8.2 Final Project: Design and Build a Database frame and live demo: rules across rows, studio: Milestone 5, Business logic, studio: Milestones 6 and 7, Security and backup
1:30–1:40	*Break*
1:40–2:15	8.2 Final Project: Design and Build a Database showcase
2:15–2:50	8.3 Next Steps & Best Practices frame and live demo, start the lesson project
2:50–3:00	Wrap-up and course close

📅 SAMPLE SCHEDULES

No dates: fit them to your calendar. "Twice weekly" puts two sessions in each week (Day 1 and Day 2), finishing in week 7.

SESSION	TWICE WEEKLY (≈7 WEEKS)	WEEKLY (14 WEEKS)
1 What Databases Are and Installing MySQL	Week 1, Day 1	Week 1
2 The MySQL Client and Your First Tables	Week 1, Day 2	Week 2
3 Data Types, Constraints and CRUD	Week 2, Day 1	Week 3
4 Filtering, Sorting and Limiting	Week 2, Day 2	Week 4
5 Aggregates and Subqueries	Week 3, Day 1	Week 5

SESSION	TWICE WEEKLY (≈7 WEEKS)	WEEKLY (14 WEEKS)
6 Relationships and Foreign Keys	Week 3, Day 2	Week 6
7 Joins and the Part 1 Assessment	Week 4, Day 1	Week 7
8 CTEs and Window Functions	Week 4, Day 2	Week 8
9 Indexes and Views	Week 5, Day 1	Week 9
10 Normalization and Stored Programs	Week 5, Day 2	Week 10
11 Transactions and JSON	Week 6, Day 1	Week 11
12 Users, Permissions and Backups	Week 6, Day 2	Week 12
13 Final Project Kickoff and Studio I	Week 7, Day 1	Week 13
14 Final Project Studio II, Showcase and Next Steps	Week 7, Day 2	Week 14

🏁 CHECKPOINTS, ASSESSMENT AND PROJECT

- **Self-checks and exit tickets** in every session are practice, not grades: they tell you and your instructor what to revisit.
- **Checkpoint 1 (Session 4)** and **Checkpoint 2 (Session 11)**, 30 minutes each: short individual tasks on the modules just finished, scored with a short rubric.
- **Part 1 assessment (Session 7):** queries and a small design task on the bookstore database, scored with a rubric. When Part 1 runs on its own, it is the course's final assessment.
- **Final project (Sessions 12 to 14).** From a short brief (a neighborhood tool library), you design the tables, keys and rules, build and load the database, write the queries and views it needs, give an application account only the permissions it needs, back it up, and present it. A well-reasoned modest design scores as well as an elaborate one.

🧰 REQUIRED SOFTWARE AND DATA

- **MySQL 9.7 LTS:** Community Server: Windows MSI, macOS Homebrew or DMG, Linux APT/Yum repository, or Docker `mysql:9.7` (MySQL 8.4 and MariaDB work with noted differences)
- **The mysql command-line client:** installed with the server; optional graphical tools such as phpMyAdmin or MySQL Workbench
- **A text editor:** for .sql files (VS Code is free and works well)
- **The course database:** `bookstore.sql` from the course site (built step by step in Part 1, reloaded from Lesson 4.2 on)
- **No accounts or paid tools:** everything runs on an ordinary laptop

MySQL installs once, at home, before Session 1, following Lesson 1.2. Your instructor sends a setup guide and an environment check a week ahead; Session 1 has an install clinic for anything that went wrong.

✉️ FOR INSTRUCTORS

Want to teach this course? A complete instructor kit for MySQL Foundations (facilitator guides, slides, labs, workbooks, answer keys and assessments for all 14 sessions) is available under license. Ask about the instructor kit.